Data-Driven Sales and Inventory Optimization for a Rural Kirana Store

A MidTerm report for the BDM capstone Project

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1. Executive Summary

Bhavana Provision General Store is a rural kirana shop located in Pimpri Khurd, Maharashtra, and has been operating since 2015. The store sells essential household grocery items such as cooking oil, tea, pulses, biscuits, and packaged food products to customers from nearby villages. Although the store experiences regular daily sales, most decisions related to inventory management and customer credit are based on manual records and the shop owner's experience. Because there is no structured analysis of transaction data, it becomes difficult to identify which products contribute the most revenue, which items are fast-moving, and how customer credit affects the store's cash flow and profitability.

To study these issues, 6465 transaction records from the store were digitized and organized for analysis. The dataset contains important variables such as Date, ItemName, Quantity, PricePerUnit, TotalAmount, PaymentMode, AmountPaid, PendingDues, and CustomerName, which capture details about product sales, payment behavior, and transaction patterns. Descriptive statistical analysis shows that the store generated total sales of ₹6,34,657, with an average transaction value of ₹98.17 and a median value of ₹50. The dataset also indicates that ₹3,76,798 has been received as payments, while ₹2,57,859 remains pending, highlighting the role of credit transactions in the business.

For the analysis, Microsoft Excel tools such as Pivot Tables, sorting, aggregation formulas, and visual charts were used to examine product demand patterns, revenue distribution, and customer payment behavior. The results show that products such as Groundnut Oil, Sunflower Oil, and Tur Dal contribute the highest share of revenue. At the same time, items like Red Label Tea and Ghadi Detergent Cake appear as fast-moving products based on quantity sold. These findings highlight the importance of structured data analysis to support better inventory planning and improved monitoring of customer credit transactions.

2. Proof of Originality of Data

To confirm that the data used in this project is original and collected from the actual business, the following evidence has been provided. These materials show that the analysis is based on real transaction records from the store.

2.1 Video Interaction with Store Owner

A short video interaction was conducted with the store owner, Mr. Sunil Surve. In the video, he explains the background of the shop, daily business activities, credit practices, and some operational challenges faced in managing the store. This interaction confirms that the project is based on direct communication with the business owner.

2.2 Images of the Store Premises

Photographs of the shop have been taken during the visit. These images show the shop environment, product shelves, billing area, and the place where daily sales records are maintained.

2.3 Sales Register Records

The store maintains handwritten registers where daily sales and credit transactions are recorded. Some pages of these registers have been scanned or photographed and attached as proof of the original records.

2.4 Digitized Transaction Dataset

The handwritten sales records were later converted into a digital dataset using Excel. The dataset includes important columns such as Date, ItemName, Quantity, PricePerUnit, TotalAmount, PaymentMode, AmountPaid, PendingDues, and CustomerName. This dataset represents actual transactions recorded in the store.

2.5 Confirmation Letter from Store Owner

A signed letter from the store owner has been attached, confirming that permission was given to use the store's data for this academic project.

2.6 Evidence Folder Link

All supporting materials including the video interaction, store images, scanned register pages, confirmation letter, and dataset file are available in the shared Google Drive folder below:

https://drive.google.com/drive/folders/116h5_MviqgJ0XKVU8xHpdkw5FaKSRN7j?usp=drive_link

Figure 1. Meet with owner

Figure 2. bhavana provision general store

3. Metadata

3.1 Definition of Metadata

Metadata means **“**data about the data**”**. It explains the information collected in the dataset, including the meaning of each variable, the type of data, and why that variable is important for the analysis. In this project, metadata helps to understand how the collected transaction data represents the business activities of Bhavana Provision General Store. By clearly defining the variables used in the dataset, it becomes easier to analyze product sales, customer payment behaviour, and inventory demand patterns.

3.2 Description of Variables Collected

The dataset used in this project contains transaction-level information collected from the store’s sales registers. Each row represents one sales transaction recorded at the shop. The main variables included in the dataset are described below.

Variable	Description	Importance
Date	The date on which the transaction took place	Helps analyze sales trends over time
ItemName	Name of the product sold	Used to identify product demand and performance
Quantity	Number of units sold in the transaction	Helps identify fast-moving products
PricePerUnit	Price of one unit of the product	Used to calculate total revenue
TotalAmount	Total value of the transaction	Important for revenue analysis

PaymentMode	Type of payment (Cash or Credit)	Helps analyze customer payment behaviour
AmountPaid	Amount paid by the customer at the time of purchase	Used to track actual cash inflow
PendingDues	Remaining unpaid amount	Helps identify credit transactions and outstanding balances
CustomerName	Name of the customer	Useful for studying customer purchase patterns

3.3 Relevance of Variables to the Problem Statement

The selected variables are directly related to the problems identified in the store's operations. One major issue in the store is the lack of structured analysis of sales data, which makes it difficult to understand product demand and inventory requirements. Variables such as ItemName, Quantity, and TotalAmount help identify high-demand and high-revenue products, which is useful for better inventory planning.

Another important challenge is customer credit management. Many transactions in rural kirana stores are done on credit, which affects cash flow. Variables such as PaymentMode, AmountPaid, and PendingDues help analyze credit transactions and understand how much payment is still outstanding.

By organizing and analyzing these variables, the dataset provides useful insights into product performance, sales patterns, and customer payment behaviour. This information helps support data-driven decision-making for improving inventory management and monitoring customer credit in the store.

Date	InvoiceID	CustomerName	ItemName	Category	Quantity	PricePerUnit	TotalAmount	PaymentMode	AmountPaid	PendingDues	Month
2025-10-01	INV-20251001-367	Radha Tai	Santoor Soap (Small)	Personal Care	1	35	35	Credit	3	32	Oct
2025-10-01	INV-20251001-367	Radha Tai	Wheel Powder (1kg)	Household	1	65	65	Credit	0	65	Oct
2025-10-01	INV-20251001-755	Narayan	Vim Bar (Green)	Household	4	10	40	Cash	40	0	Oct
2025-10-01	INV-20251001-755	Narayan	Balaji Wafers / Chivda	Snacks	4	10	40	Cash	40	0	Oct
2025-10-01	INV-20251001-572	Namdev Baba	Stationery Notebook	Others	2	15	30	Cash	30	0	Oct
2025-10-01	INV-20251001-572	Namdev Baba	Colgate Toothpaste (Small)	Personal Care	5	20	100	Cash	100	0	Oct
2025-10-01	INV-20251001-572	Namdev Baba	Ghadi Detergent Cake	Household	5	10	50	Cash	50	0	Oct
2025-10-01	INV-20251001-572	Namdev Baba	Rice (HMT/Kolam)	Staples	0.25	55	14	Cash	14	0	Oct
2025-10-01	INV-20251001-752	Jyoti Tai	Rice (HMT/Kolam)	Staples	5	55	275	Cash	275	0	Oct
2025-10-01	INV-20251001-752	Jyoti Tai	Godrej No. 1 Soap	Personal Care	3	30	90	Cash	90	0	Oct
2025-10-01	INV-20251001-752	Jyoti Tai	Sunsilk Shampoo (Sachet)	Personal Care	3	1	3	Cash	3	0	Oct
2025-10-01	INV-20251001-246	Bajirao	Balaji Wafers / Chivda	Snacks	2	10	20	Credit	4	16	Oct
2025-10-01	INV-20251001-246	Bajirao	Stationery Notebook	Others	1	15	15	Credit	0	15	Oct
2025-10-01	INV-20251001-321	Narayan	Blue Ball Pen	Others	4	5	20	Cash	20	0	Oct
2025-10-01	INV-20251001-782	Dada Patil	Sunflower Oil (1L Packet)	Staples	2	135	270	Cash	270	0	Oct
2025-10-01	INV-20251001-539	Kamla Bai	Mobile Recharge Top-up	Others	5	50	250	Cash	250	0	Oct

Figure 3.data overview

4. Descriptive Statistics

Descriptive statistics are used to summarize and understand the main characteristics of the collected dataset. These statistics help in explaining the overall pattern of sales transactions in Bhavana Provision General Store. By calculating measures such as mean, median, mode, and standard deviation, it becomes easier to understand the average transaction behaviour, variation in product sales, and customer payment patterns.

In this project, descriptive statistics were calculated for key numerical variables such as Quantity, TotalAmount, AmountPaid, and PendingDues. The mean (average) value helps in understanding the typical level of sales and payment amounts recorded in the store. For example, the average value of TotalAmount indicates the usual size of a customer transaction. The median represents the middle value when the data is arranged in ascending order, which helps identify the central tendency of the dataset without being affected by extreme values.

The mode identifies the most frequently occurring value in the dataset. This can help identify common purchase quantities or frequently occurring transaction values. The standard deviation measures the variation in the dataset and shows how much the values differ from the average. A higher standard deviation indicates that transaction amounts vary significantly across customers and products.

These summary statistics provide an overview of the store's sales and payment behaviour. For example, statistics related to Quantity and TotalAmount help identify typical product demand and transaction size, while statistics related to AmountPaid and PendingDues help understand customer credit behaviour. This information is directly related to the problem statement of the project, which focuses on improving product demand understanding, monitoring credit transactions, and supporting better decision-making for inventory and financial management.

Statistic	Quantity	PricePerUnit	TotalAmount	AmountPaid	PendingDues
count	6465	6465	6465	6465	6465
mean	3	38	98	58	40
std	2	38	130	111	92
min	0	1	1	0	0
25%	1	10	22	0	0
50%	3	30	50	18	0
75%	4	45	125	67	40
max	5	175	875	875	875

Figure 4.descriptive statistics

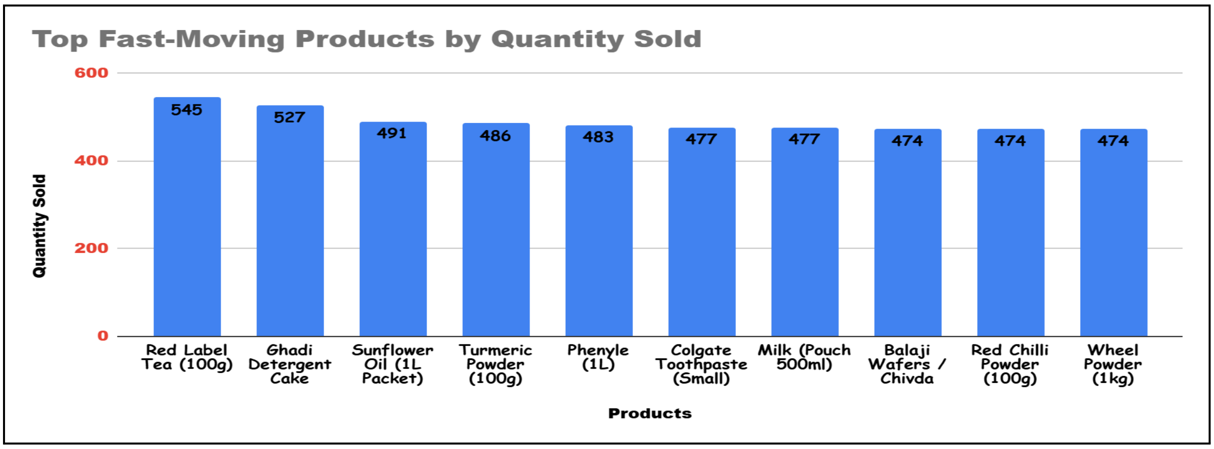


Figure 5.Fast-Moving Products Based on Quantity Sold

Analysis: This chart shows the quantity of products sold in the store. Items such as Red Label Tea, Ghadi Detergent Cake, and Sunflower Oil appear among the highest sold products, indicating strong customer demand.

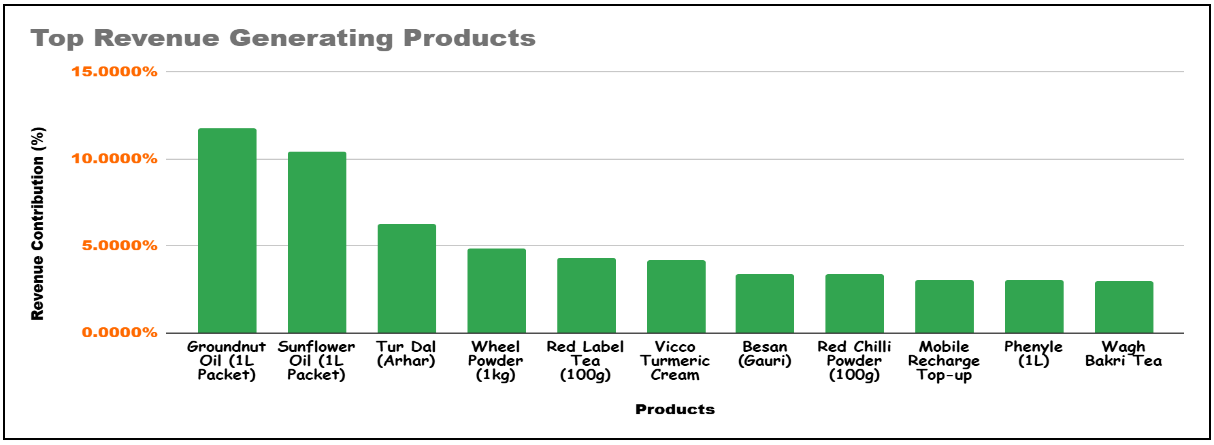


Figure 6: Revenue Contribution by Top Products

5. Detailed Explanation of Analysis Process / Method

5.1 Data Cleaning and Preprocessing

Before starting the analysis, the collected data was cleaned and prepared to ensure that it is accurate and suitable for analysis. The sales data of Bhavana Provision General Store was originally recorded in handwritten registers maintained by the store owner. During the store visit, I discussed the store operations and sales records with the shop owner and collected the transaction details from these registers. The records were then manually entered into an Excel sheet to create a digital dataset for further analysis.

While converting the handwritten records into a digital format, several data cleaning steps were required. In many cases, product names were written differently in the registers due to spelling variations or abbreviations. For example, the same product might appear as “Sunflower Oil”, “Sunflower Oil 1L”, or “Sunflower Oil Pack”. Such variations were standardized to maintain consistency in the dataset. Similarly, spelling errors that occurred during manual data entry were corrected to ensure that the same product was not counted as different items.

Some records also contained incomplete or inconsistent values. These entries were carefully reviewed and corrected wherever possible using information from the original register entries. Duplicate records were checked and removed to avoid double counting of transactions. Numerical values such as Quantity, PricePerUnit, TotalAmount, AmountPaid, and PendingDues were verified to ensure that they were stored in the correct numeric format for analysis.

These data cleaning and preprocessing steps were necessary to ensure that the dataset is reliable and ready for analysis. Proper cleaning of the data helps avoid errors in calculations and improves the accuracy of the results obtained from statistical analysis and visualization.

5.2 Analysis Process and Methods

After cleaning the dataset, several analytical methods were used to understand the sales patterns, product demand, and customer payment behaviour in the store. The analysis process was carried out using Microsoft Excel tools such as Pivot Tables, sorting functions, aggregation formulas, and charts. The main analysis steps are described below.

1. Product Revenue Analysis

To identify which products generate the highest revenue for the store, a pivot table analysis was performed. In this method, the total sales value for each product was calculated by aggregating transaction values.

Mathematically, the revenue of a product is calculated as:

Revenue = Quantity × PricePerUnit

Using pivot tables, the SUM of TotalAmount was calculated for each product. This method helps identify high-revenue products and understand revenue concentration in the store.

This analysis directly supports the project problem, which focuses on identifying key products that contribute most to the store’s sales.

2. Product Demand Analysis

Product demand was analyzed by calculating the total quantity sold for each product. The quantity values from all transactions were aggregated to determine which items are sold most frequently.

This method helps identify fast-moving products, which are items that sell in higher quantities and require frequent restocking.

Understanding product demand helps improve inventory planning and ensures that high-demand items remain available in the store.

3. Customer Payment and Credit Analysis

Another important part of the analysis was studying customer payment behaviour. For this purpose, the variables AmountPaid and PendingDues were analyzed.

This analysis helps determine:

- 3.1 how much payment is received immediately
- 3.2 how much amount remains unpaid as credit

By comparing these values across customers and transactions, it becomes possible to understand the level of credit usage in the store.

This analysis is directly connected to the problem of credit-based transactions affecting cash flow in the business.

4. Payment Mode Distribution Analysis

The dataset also contains a variable called PaymentMode, which indicates whether the transaction was made using cash or credit.

A frequency analysis was performed to calculate the number of transactions under each payment mode. This helps understand the overall payment behaviour of customers.

This analysis provides insights into how much the store depends on credit transactions and how it may impact short-term cash flow.

5. Time-Based Sales Analysis

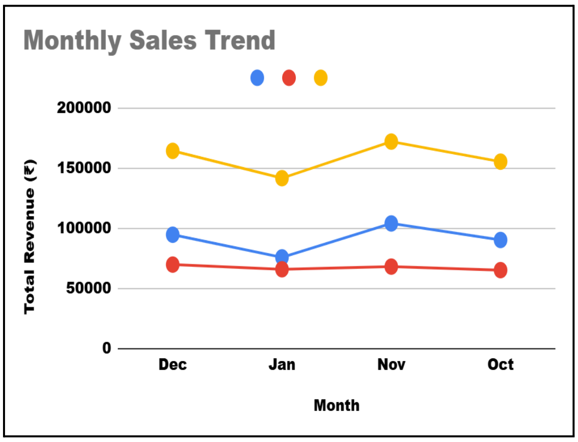
To understand sales patterns over time, the transaction dates were grouped by month. Total sales for each month were calculated using aggregation methods.

This method helps identify whether sales increase or decrease over different time periods and supports better planning of inventory and stock management.

6. Results and Findings

This section presents the main findings obtained from the analysis of the kirana store's transaction data. Different charts and visualizations are used to identify patterns in product demand, revenue distribution, customer purchase behaviour, and payment modes. These insights help understand how the store's sales are distributed across products, customers, and time.

6.1 Payment Mode Distribution



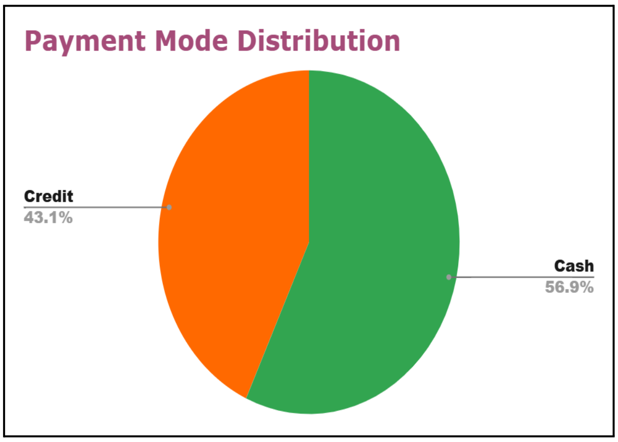


Figure 7: Payment Mode Distribution *Figure 8: Monthly Sales Trend*

Analysis: Figure 3 shows the distribution of payment modes used by customers. Cash transactions account for the majority of sales (57%), while credit transactions represent 43% of the total transactions. This indicates that credit sales form a significant portion of the store's business and may affect short-term cash flow if payments are delayed.

Figure 5 shows the monthly sales trend of the store. The chart indicates that sales vary across different months, with slightly higher revenue observed in certain months. These variations may be influenced by seasonal demand, customer purchasing patterns, or local factors affecting sales.

6.2 Customer Credit and Pending Dues

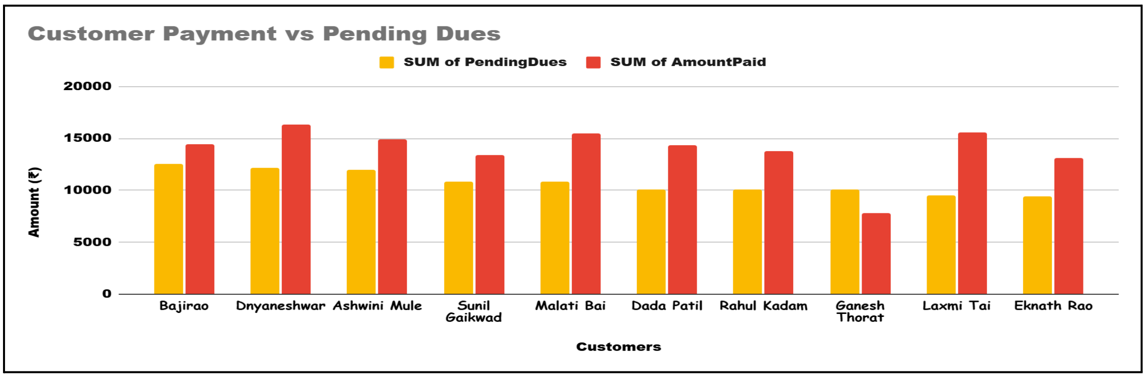


Figure 9: Customer Payment vs Pending Dues

Analysis : The chart compares the amount paid by customers with their pending dues. It shows that while many customers make payments regularly, some still have outstanding balances, highlighting the importance of tracking credit transactions.

6.3 Customer Purchase Frequency

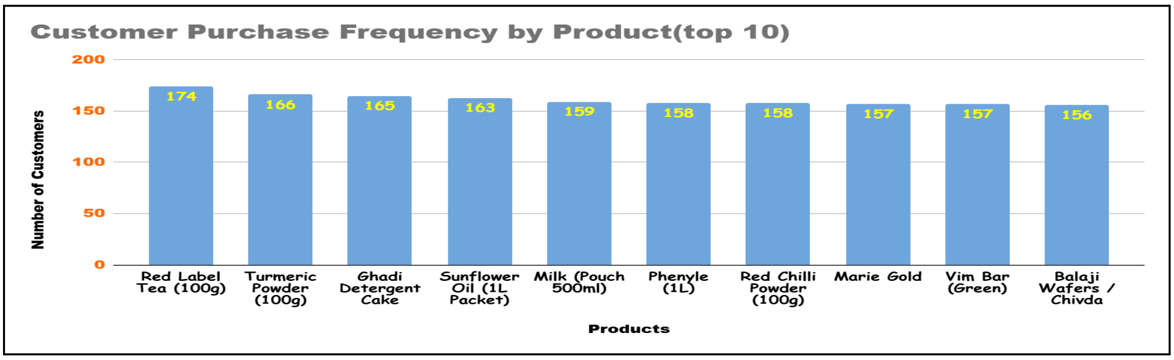


Figure 10: Customer Purchase Frequency by Product

Analysis : This chart shows how frequently different products appear in customer transactions. Items such as Red Label Tea (100g), Turmeric Powder (100g), and Ghadi Detergent Cake have the highest purchase frequency among customers.

This indicates that these products are regularly bought by customers and represent essential daily-use items in the store.

6.4 Revenue Distribution by Product Category

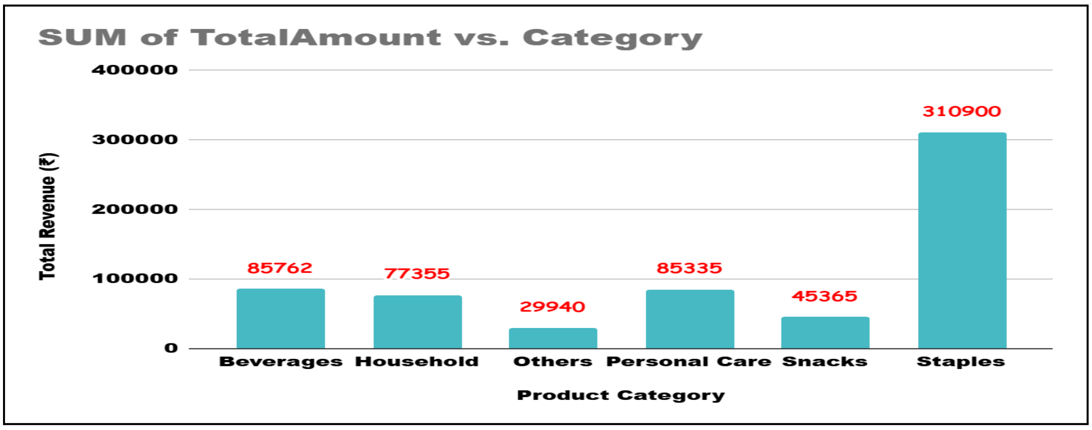


Figure 11: Revenue Distribution by Product Category

Analysis : This chart shows the total revenue generated from different product categories in the store. The Staples category contributes the highest revenue (₹310,900) compared to other categories such as beverages, household, snacks, and personal care.

This indicates that essential grocery items such as grains, pulses, and cooking ingredients form the core of the store's sales and play a major role in overall revenue generation.